

Split shifts

Very few animals stay active around the clock. Instead, most are either active by day (diurnal) or active by night (nocturnal). By waking at different times, diurnal and nocturnal animals can live in similar ways without competing head-on for food. For example, butterflies and moths both feed at flowers, but butterflies come out during the day, while most moths are active after dark.

In the same way, raptors, or birds of prey, hunt by day—at night, they are replaced by owls.



Feeding with the tides

For sanderlings and many other waders, the rise and fall of the tide is much more important than the rhythm of night and day. When the tide is low, sanderlings scuttle along the water's edge, snapping up small animals that are exposed by the waves. At high tide, they stop feeding and rest until the tide turns once more. Waders have very sensitive bills, and they often feed by moonlight when the tide is going out.

Butterfly

Hawk moth

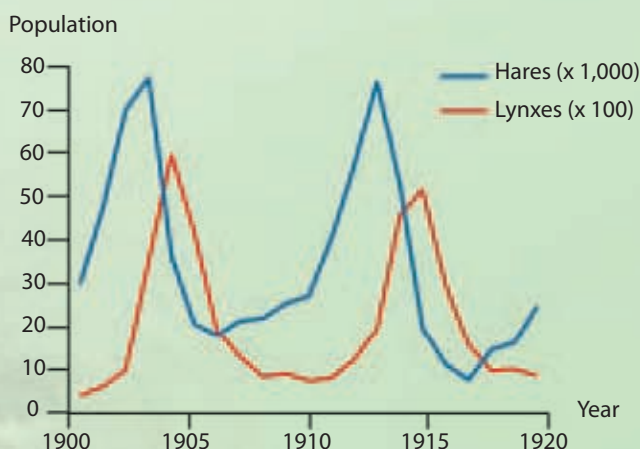
Butterfly lands on flowers when it feeds, holding on tightly with its legs

Hawk moth hovers in front of flowers, keeping its legs tucked up against its body

Outer covering, or exoskeleton, is shed for a final time before cicada mates and lays eggs

Population cycles

Animal numbers often change from year to year. These changes are caused by many factors, including the weather and food supply. Usually, they are impossible to predict, but in some animals, they rise and fall in cycles that last a set number of years.



Canadian lynx with snowshoe hare prey

Boom and bust

In the Arctic, the numbers of snowshoe hares and Canadian lynxes soar and then crash every nine or 10 years. The crashes are triggered when the hares start to run short of food. The lynx population drops a year or two later, when they start to run out of hares to eat.

Years in the making

North American periodical cicadas have some of the longest rhythms in the animal world. There are several species, and each one has a life cycle lasting exactly 13 or 17 years. The cicadas spend most of their lives underground, sucking sap from roots, but after 13 or 17 years, they crawl up tree trunks, all at once, to mate and lay their eggs. For a few weeks, the air is filled with ear-piercing sounds, as billions of cicadas call to each other among the trees. After breeding, the adults die, and another 13 or 17 years pass before the next brood emerges from the ground.

